

SELF STUDY

MongoDb

1. Practice CRUD operations on Mongo.

Q1,2..

```
db.createCollection("emp")
{ ok: 1 }
db.createCollection("dept")
{ ok: 1 }
```

Q3.

```
> db.emp.updateMany(
  {deptno: 10},
  {$inc: {sal: 100}}
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

Q4.

```
> db.emp.find(
  {sal: {$gt: 1000}},
  {_id: 0, ename: 1, job: 1, deptno: 1, sal: 1}
).sort(
  {deptno: 1, job: 1, sal: -1}
)
< {
  ename: 'MILLER',
  job: 'CLERK',
  sal: 1400,
  deptno: 10
}
{
  ename: 'CLARK',
  job: 'MANAGER',
  sal: 2550,
  deptno: 10
}
{
  ename: 'KING',
  job: 'PRESIDENT',
  sal: 5100,
  deptno: 10
}
{
  ename: 'SCOTT',
  job: 'ANALYST',
  sal: 3000,
  deptno: 10
}
```

Q5.

```
db.dept.deleteOne({deptno: 40})

{
  acknowledged: true,
  deletedCount: 1
}
```

2.Learn grouping operations using Mongo aggregation pipeline

Q1.

```
> db.emp.aggregate([
  {
    $match: {
      job: {$ne: "PRESIDENT"}
    }
  },
  {
    $group: {
      _id: "$job",
      total_sal: {$sum: "$sal"},
      max_sal: {$max: "$sal"},
      min_sal: {$min: "$sal"},
      avg_sal: {$avg: "$sal"}
    }
  }
])
< {
  _id: 'ANALYST',
  total_sal: 6000,
  max_sal: 3000,
  min_sal: 3000,
  avg_sal: 3000
}
{
  _id: 'SALESMAN',
  total_sal: 5600,
  max_sal: 1600,
  min_sal: 1250,
```

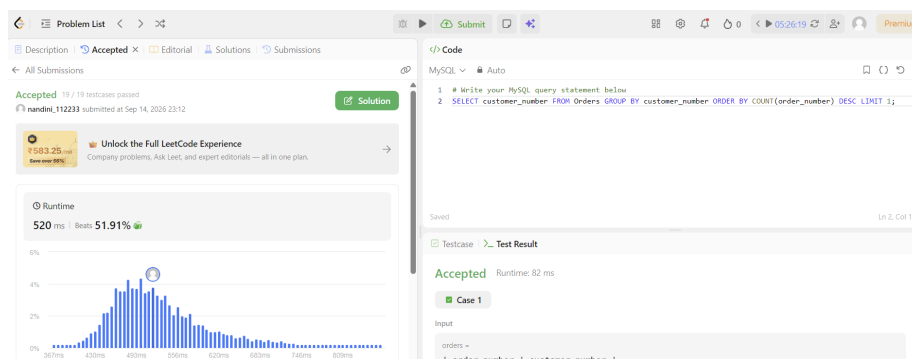
Q2.

```
db.emp.aggregate([
  {
    $lookup: {
      from: "dept",
      localField: "deptno",
      foreignField: "deptno",
      as: "dept_info"
    }
  },
  {
    $unwind: "$dept_info"
  },
  {
    $project: {
      _id: 0,
      ename: 1,
      dname: "$dept_info.dname",
      income: {
        $add: [
          "$sal",
          {$ifNull: ["$comm", 0]}
        ]
      }
    }
  },
  {
    $sort: {
      income: -1
    }
  }
])

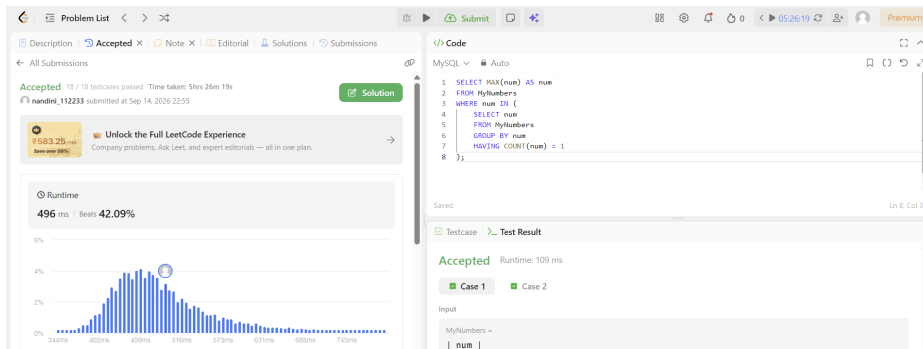
{
  ename: 'KING',
  dname: 'ACCOUNTING',
  income: 5100
}
{
  ename: 'SCOTT',
  dname: 'RESEARCH',
  income: 3000
}
```

GROUP BY Problems

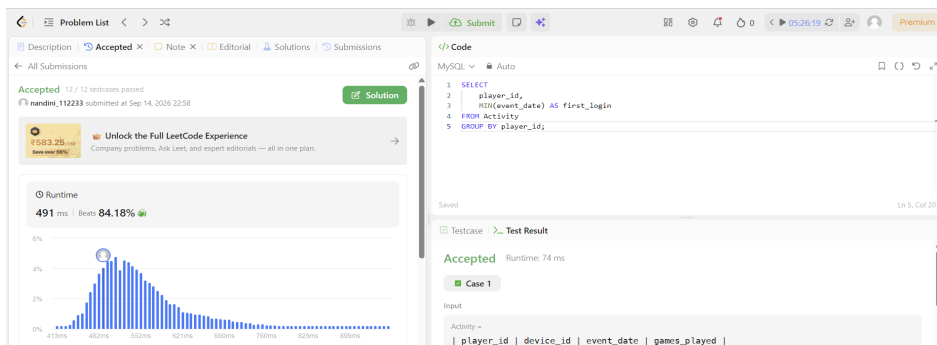
Q1.



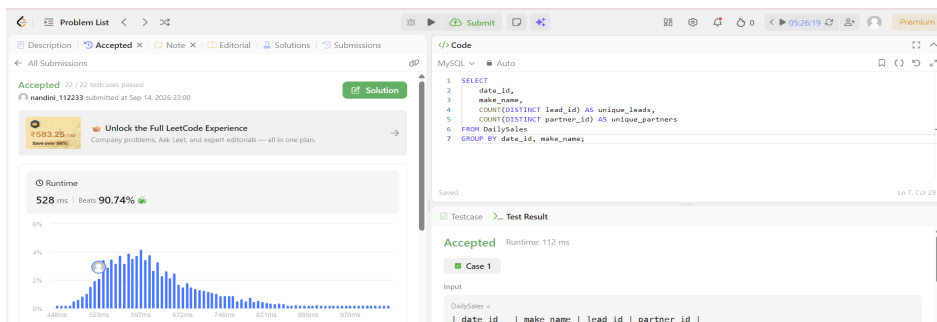
Q2.



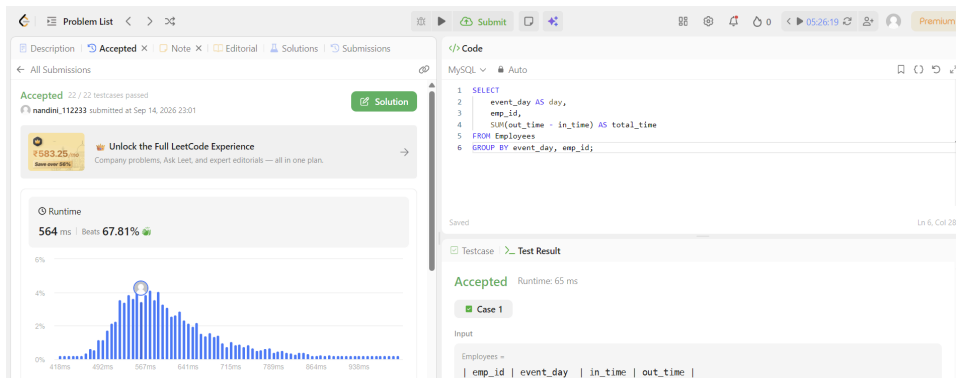
Q3.



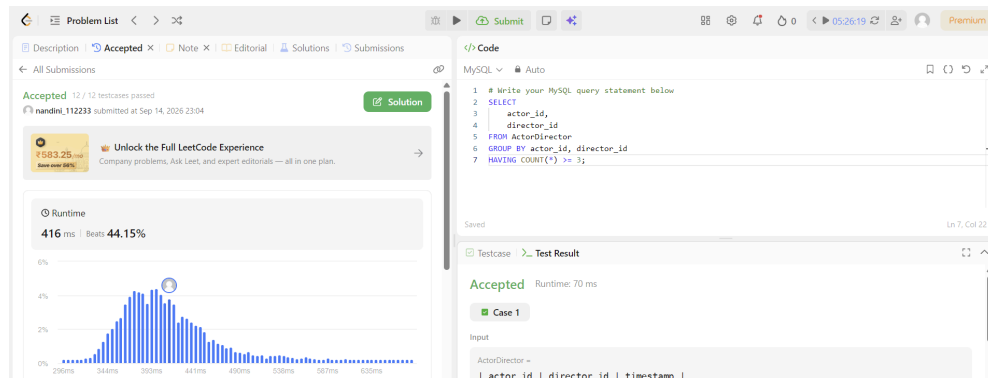
Q4.



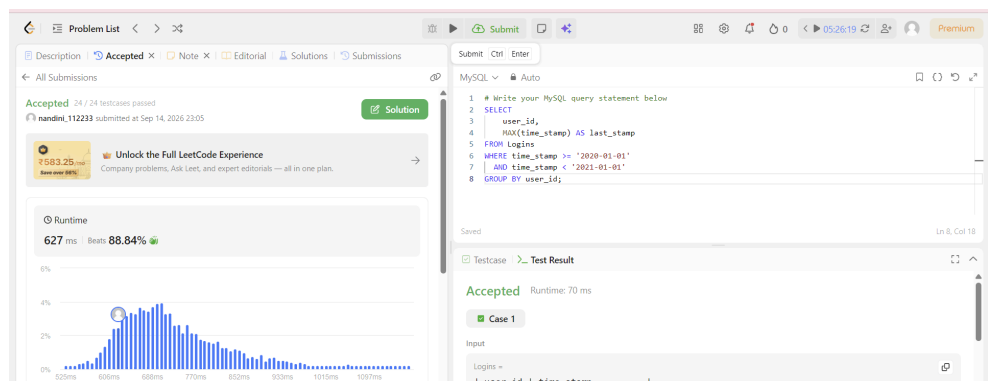
Q5.



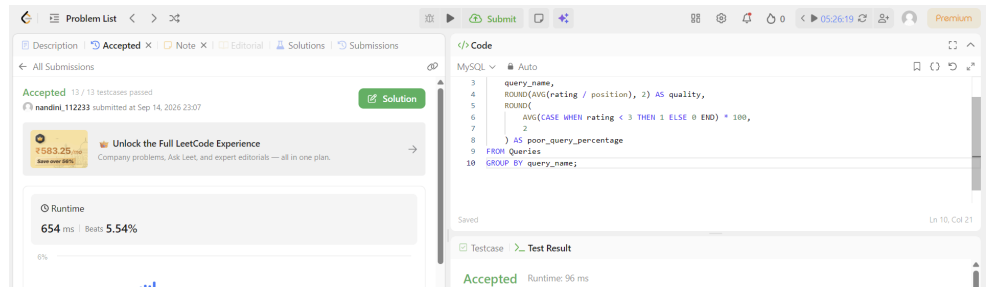
Q6.



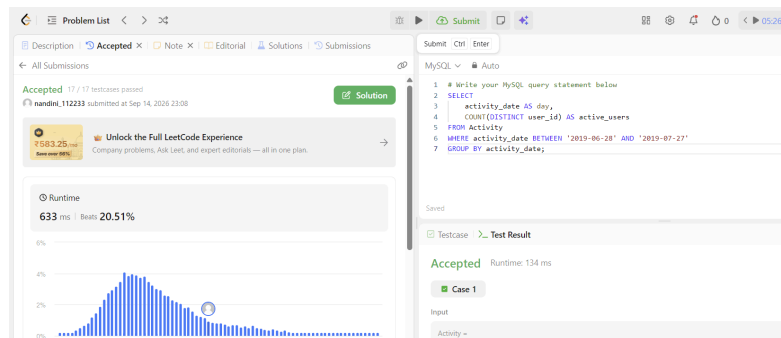
Q7.



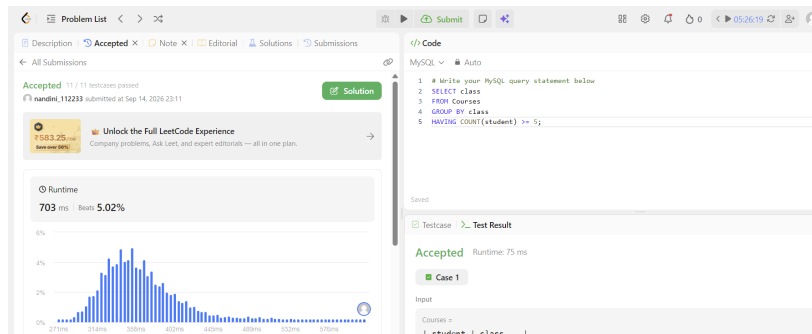
Q8.



Q9.

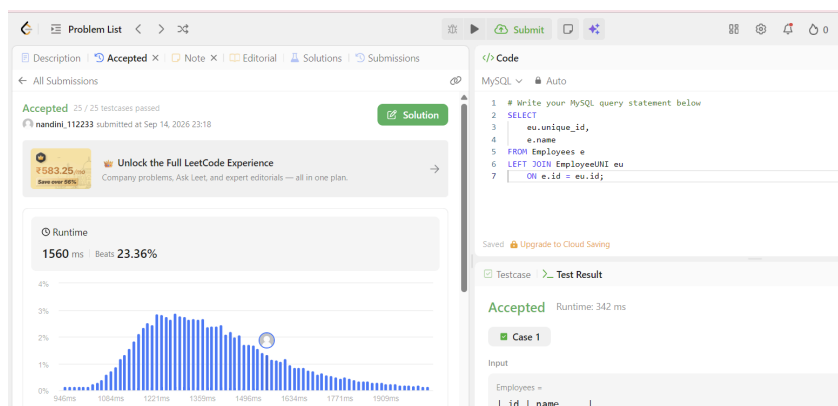


Q10.

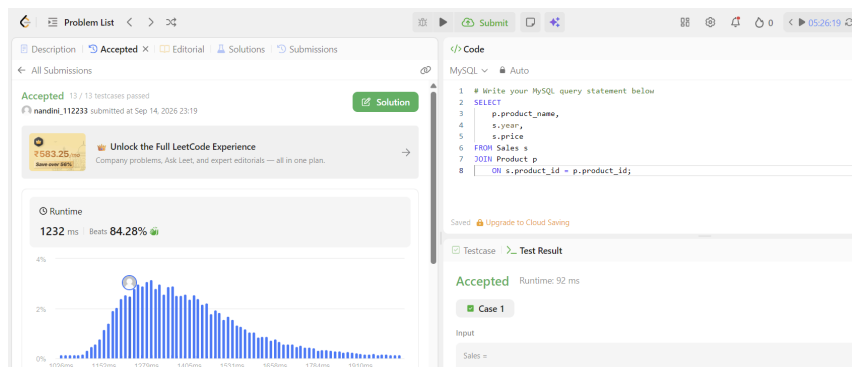


JOINS

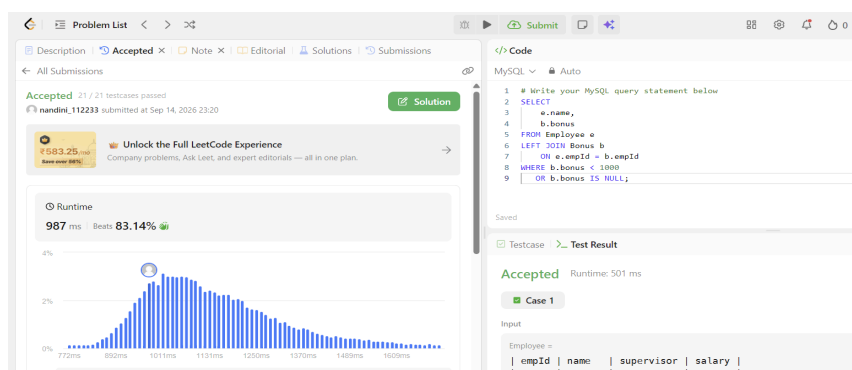
Q1.



Q2.



Q3.



Q4.

Accepted 15 / 15 testcases passed
nandini_112233 submitted at Sep 14, 2026 23:20

Runtime: 1033 ms | Beats 76.27%

Code:

```
9 LEFT JOIN Examinations e
10 ON s.student_id = e.student_id
11 AND sub.subject_name = e.subject_name
12 GROUP BY
13 s.student_id,
14 s.student_name,
15 sub.subject_name
16 ORDER BY
17 s.student_id,
18 sub.subject_name;
```

Testcase: Accepted Runtime: 171 ms

Case 1

Input:

Students =

student id | student name |

Q5.

Accepted 11 / 11 testcases passed
nandini_112233 submitted at Sep 14, 2026 23:21

Runtime: 453 ms | Beats 18.06%

Code:

```
1 # Write your MySQL query statement below
2 SELECT
3     e.name
4 FROM Employee e
5 JOIN Employee m
6 ON e.managerId = m.id
7 GROUP BY
8     m.id,
9     m.name
10 HAVING COUNT(e.id) >= 5;
```

Testcase: Accepted Runtime: 81 ms

Case 1

Input:

Employee =

id	name	department	managerId
1	Joe	IT	
2	Helen	Sales	3
3	Sam	Sales	3
4	Max	IT	3
5	Ken	IT	1
6	Jane	IT	3
7	John	Sales	3
8	Donna	IT	3
9	Michael	IT	3
10	Jay	IT	1
11	Anthony	IT	3

Q6.

HackerRank | Prepare | Certify | Compete

Prepare > SQL > Basic Join > Population Census > Submissions

Population Census

70 more points to get your first star!

Rank: 3060678 | Points: 10/80

Problem | Submissions | Leaderboard | Discussions

RESULT	SCORE	LANGUAGE	TIME	
Accepted	10.0	MySQL	a minute ago	View Results
Accepted	10.0	MySQL	2 minutes ago	View Results

NEED HELP?

- View discussions
- View top submissions

Q7.

HackerRank | Prepare | Certify | Compete

Prepare > SQL > Basic Join > Average Population of Each Continent > Submissions

Average Population of Each Continent

60 more points to get your first star!

Rank: 2963586 | Points: 20/80

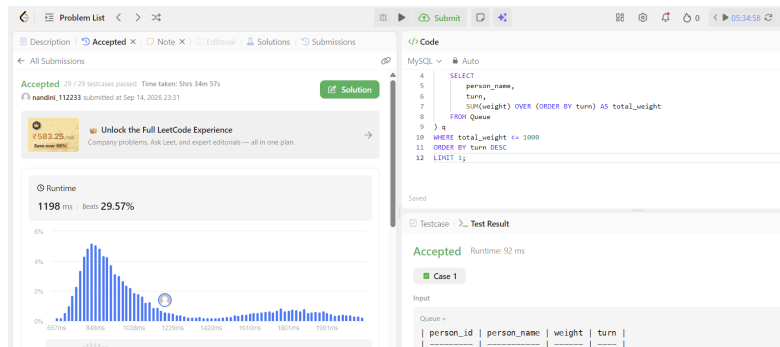
Problem | Submissions | Leaderboard | Discussions

RESULT	SCORE	LANGUAGE	TIME	
Accepted	10.0	MySQL	a few seconds ago	View Results

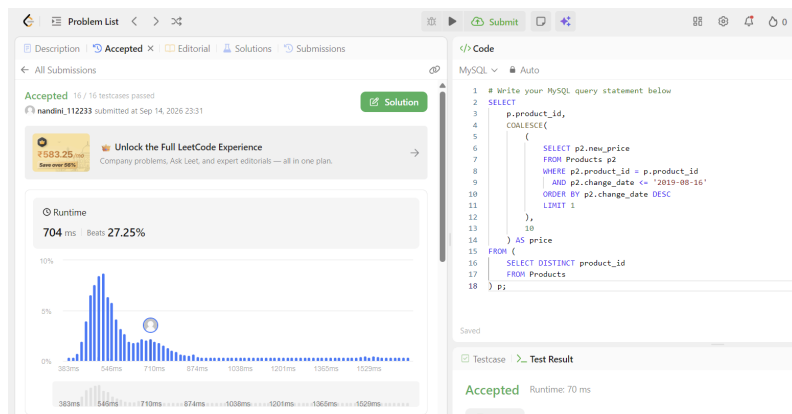
NEED HELP?

- View discussions
- View top submissions

Q8.

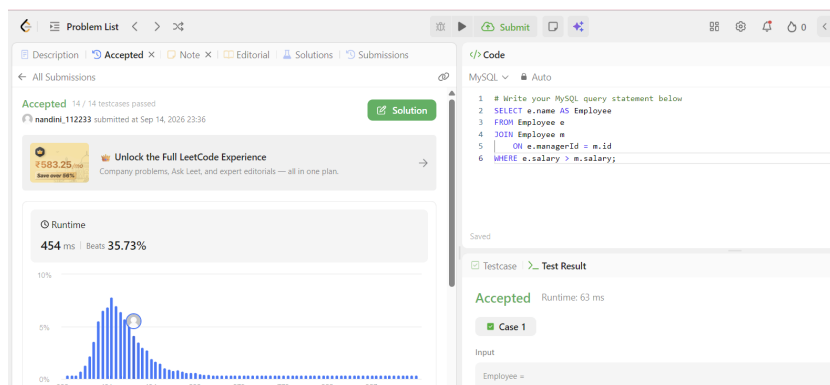


Q9.

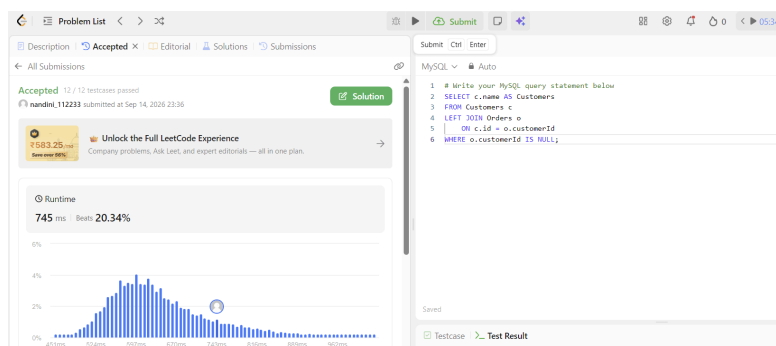


Subquery

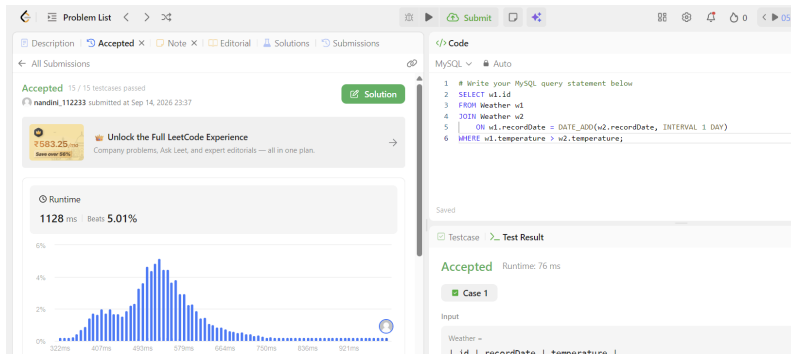
Q1.



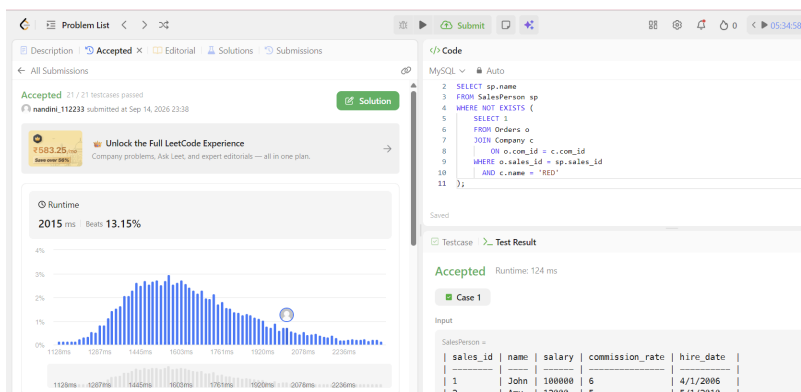
Q2.



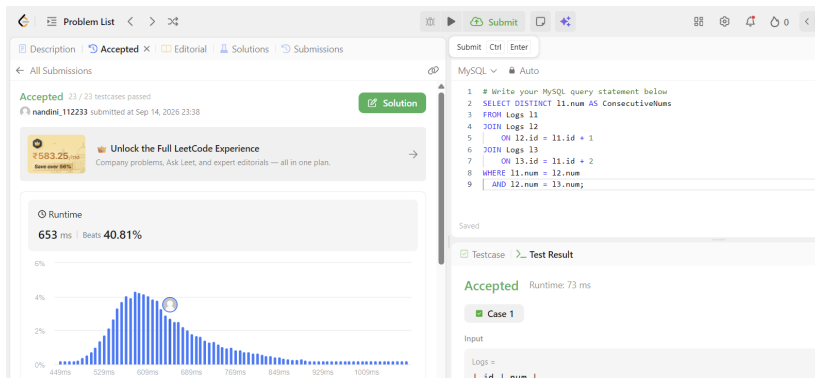
Q3.



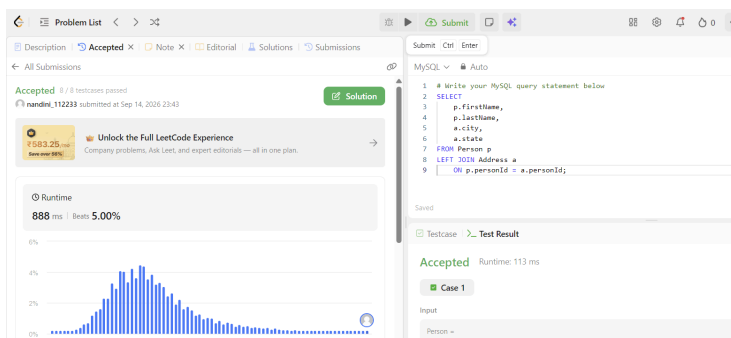
Q4.



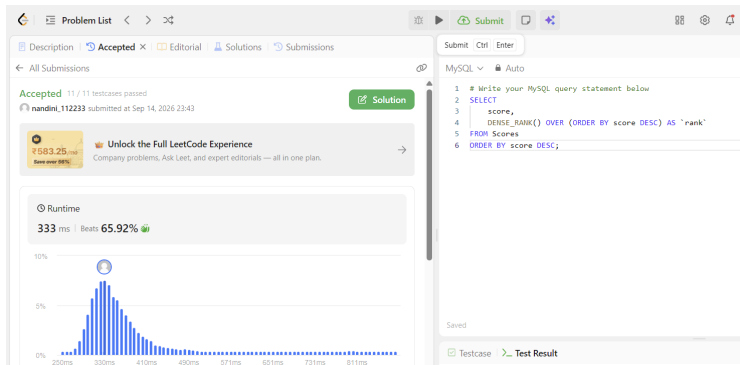
Q5.



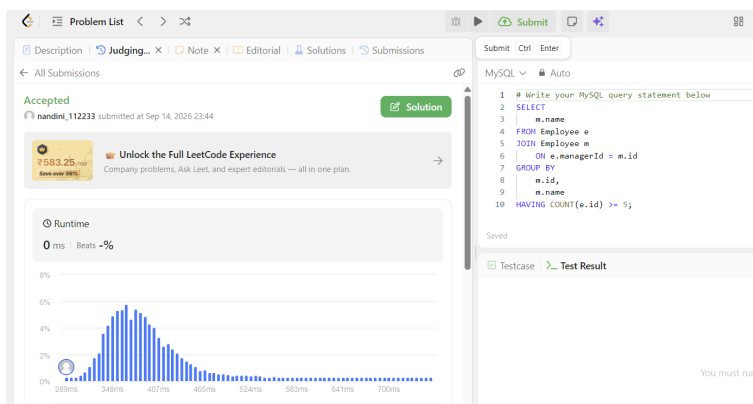
Q6.



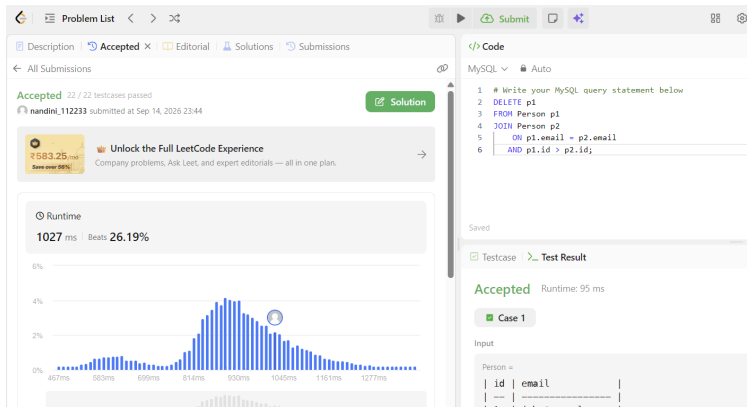
Q7.



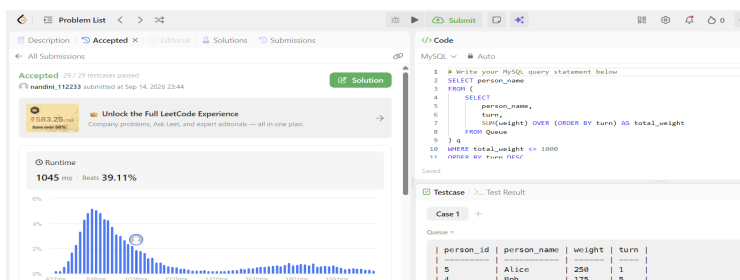
Q8.



Q9.



Q10.



Q11.

The screenshot displays the LeetCode submission interface for problem Q11. The top navigation bar includes links for Problem List, Accepted, Note, Editorial, Solutions, and Submissions. The problem status is 'Accepted' with 10/10 testcases passed, submitted by nandini_112233 on Sep 14, 2026 at 23:49. A green 'Solution' button is visible. Below the submission details, there is a promotional banner for 'Unlock the Full LeetCode Experience' with a price tag of ₹583.25. The 'Runtime' section shows a graph of execution times, with the current submission at 305 ms, beating 52.24% of other solutions. The 'Code' section shows the following MySQL query:

```
1 # Write your MySQL query statement below
2 SELECT (
3     SELECT DISTINCT salary
4     FROM Employee
5     ORDER BY salary DESC
6     LIMIT 1 OFFSET 1
7 ) AS SecondHighestSalary;
```

The 'Testcase' section shows 'Accepted' with a runtime of 105 ms. Below this, there are two test cases, 'Case 1' and 'Case 2', both marked as 'Accepted'. The 'Input' section shows the input for Case 1: Employee = [id | salary |].

CDAC Admission Process Case Study

Check if out of these tables already exist .

```
KD3-98843-Nandini@localhost> DROP TABLE IF EXISTS allocations;
Query OK, 0 rows affected, 1 warning (0.02 sec)

KD3-98843-Nandini@localhost> DROP TABLE IF EXISTS preferences;
Query OK, 0 rows affected, 1 warning (0.00 sec)

KD3-98843-Nandini@localhost> DROP TABLE IF EXISTS student_ranks;
Query OK, 0 rows affected, 1 warning (0.00 sec)

KD3-98843-Nandini@localhost> DROP TABLE IF EXISTS center_courses;
Query OK, 0 rows affected, 1 warning (0.00 sec)

KD3-98843-Nandini@localhost> DROP TABLE IF EXISTS courses;
Query OK, 0 rows affected, 1 warning (0.00 sec)

KD3-98843-Nandini@localhost> DROP TABLE IF EXISTS centers;
Query OK, 0 rows affected, 1 warning (0.00 sec)

KD3-98843-Nandini@localhost> DROP TABLE IF EXISTS students;
Query OK, 0 rows affected, 1 warning (0.02 sec)

KD3-98843-Nandini@localhost> CREATE TABLE students (
->     student_id INT PRIMARY KEY AUTO_INCREMENT,
->     name VARCHAR(100) NOT NULL,
->     email VARCHAR(100) UNIQUE NOT NULL,
->     mobile VARCHAR(15) NOT NULL
-> );
Query OK, 0 rows affected (0.05 sec)
```

Create and show tables.

```
KD3-98843-Nandini@localhost> SHOW TABLES;
+-----+
| Tables_in_cdac |
+-----+
| allocations    |
| center_courses |
| centers         |
| courses        |
| preferences    |
| student_ranks  |
| students       |
+-----+
7 rows in set (0.00 sec)
```

Records in a table

```
KD3-98843-Nandini@localhost> SELECT * FROM students;
+-----+
| student_id | name       | email           | mobile      |
+-----+
| 1          | Amit Sharma | amit@gmail.com  | 9876543210 |
| 2          | Priya Patel | priya@gmail.com | 9876543211 |
| 3          | Rahul Verma | rahul@gmail.com | 9876543212 |
| 4          | Sneha Reddy | sneha@gmail.com | 9876543213 |
| 5          | Vikram Singh | vikram@gmail.com | 9876543214 |
+-----+
5 rows in set (0.00 sec)

KD3-98843-Nandini@localhost> SELECT * FROM centers;
+-----+
| center_id | center_code | center_name      | city      |
+-----+
| 1          | IACSD       | IACSD Akurdi    | Pune      |
| 2          | ACTS        | CDAC ACTS Pashan | Pune      |
| 3          | VITA        | VITA Juhu       | Mumbai    |
+-----+
3 rows in set (0.00 sec)

KD3-98843-Nandini@localhost> SELECT * FROM courses;
+-----+
| course_id | course_code | course_name      |
+-----+
| 1          | DAC         | PG Diploma in Advanced Computing |
| 2          | DMC         | PG Diploma in Mobile Computing    |
+-----+
2 rows in set (0.00 sec)

KD3-98843-Nandini@localhost> SELECT * FROM center_courses;
+-----+
| center_course_id | center_id | course_id | capacity | filled_seats |
+-----+
| 1                | 1         | 1         | 2         | 0             |
| 2                | 2         | 1         | 1         | 0             |
| 3                | 3         | 1         | 2         | 0             |
| 4                | 2         | 2         | 1         | 0             |
+-----+
4 rows in set (0.00 sec)

KD3-98843-Nandini@localhost> SELECT * FROM student_ranks;
+-----+
| student_id | ccat_rank |
+-----+
| 1          | 10        |
| 2          | 25        |
| 3          | 40        |
| 4          | 55        |
| 5          | 120       |
+-----+
5 rows in set (0.00 sec)

KD3-98843-Nandini@localhost> SELECT * FROM preferences;
+-----+
| preference_id | student_id | center_course_id | preference_number |
+-----+
| 1             | 1          | 1                | 1                |
| 2             | 2          | 1                | 2                |
| 3             | 3          | 1                | 1                |
| 4             | 4          | 2                | 1                |
| 5             | 3          | 1                | 1                |
| 6             | 4          | 2                | 2                |
| 7             | 5          | 4                | 1                |
| 8             | 4          | 4                | 2                |
| 9             | 5          | 1                | 1                |
+-----+
9 rows in set (0.00 sec)
```

Process round allocation

```
Query OK, 0 rows affected (0.02 sec)

KD3-98843-Nandini@localhost>
KD3-98843-Nandini@localhost> DELIMITER ;
KD3-98843-Nandini@localhost>
KD3-98843-Nandini@localhost> CALL ProcessRound1Allocation();
```

allocation_id	name	ccat_rank	center_code	course_code	allocated_round	allocation_status
1	Amit Sharma	10	ACTS	DAC	1	ALLOCATED
2	Priya Patel	25	ACTS	DAC	1	ALLOCATED
3	Rahul Verma	40	IACSD	DAC	1	ALLOCATED
4	Sneha Reddy	55	ACTS	DMC	1	ALLOCATED

```
4 rows in set (0.09 sec)

Query OK, 0 rows affected (0.13 sec)
```

With store processor we call out the round 2 counselling

```
Query OK, 0 rows affected (0.02 sec)

KD3-98843-Nandini@localhost>
KD3-98843-Nandini@localhost> DELIMITER ;
KD3-98843-Nandini@localhost> CALL ProcessRound2Allocation();
```

allocation_id	name	ccat_rank	center_code	course_code	allocated_round	allocation_status
1	Amit Sharma	10	ACTS	DAC	2	ALLOCATED
2	Priya Patel	25	ACTS	DAC	2	ALLOCATED
3	Rahul Verma	40	IACSD	DAC	2	ALLOCATED
4	Sneha Reddy	55	ACTS	DMC	2	ALLOCATED

```
4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.05 sec)

KD3-98843-Nandini@localhost>
```